FLE TRIBE

WEEK 06 · WEDNESDAY · LESSON GUIDE

Loops: repeating work

This week: Write small programs that store values, make decisions, repeat work and reuse code with functions.

Why loops

A loop repeats code so you don't have to copy it. Use one whenever you'd otherwise write the same line again and again.

for: when you know how many times

```
for (let i = 1; i <= 5; i++) {
  console.log('Lap', i);
}
```

while: until something changes

```
let balance = 1000;
let months = 0;
while (balance > 0) {
  balance -= 150;
  months++;
}
console.log(months); // 7
```

for...of: go through a list

```
const fruits = ['mango', 'pawpaw', 'orange'];
for (const fruit of fruits) {
  console.log(fruit);
}
```

Watch out

- An infinite loop never ends and freezes the tab: make sure the condition eventually becomes false.

- `break` stops the loop early; `continue` skips to the next round.

Research questions

Answer these in the app, in your own words, to unlock the next lesson.

1. Print the numbers 1 to 100, but "Fizz" for multiples of 3 and "Buzz" for multiples of 5.
2. When would you use `while` instead of `for`?

Start your research here

- javascript.info: https://javascript.info/while-for
- MDN: https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Scripting/Loops